

Beyond Point Ranks: Confidence Sets for a Shared-Item LLM Leaderboard

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Great perfection seems incomplete, yet its use is never impaired.
— Tao Te Ching, Chapter 45

Abstract

Leaderboards turn finite-benchmark scores into exact model ranks, even when neighboring expected accuracies are weakly separated. We study rank uncertainty for 13 large language models evaluated on the same stratified sample of 2,000 MMLU-Pro questions. Following Mogstad, Romano, Shaikh, and Wilhelm (2024), hereafter MRSW, we construct marginal and simultaneous confidence sets for model ranks. Among all evaluated models, DeepSeek V3 has point rank one and marginal 95% rank set $\{1\}$. Four models cannot be excluded from the top three. These statements quantify uncertainty from sampling benchmark items from a target question population; they do not measure prompt sensitivity, model-version drift, or repeated-generation variability. Our method provides an alternative way to understand model performance comparisons.

Code and data availability. Code and per-question results: <https://github.com/jimmy90/llm-rank-confidence-sets>

1 Introduction

LLM benchmarks typically report one accuracy per model and sort those values into an exact leaderboard. This presentation is convenient, but it suppresses a basic distinction: the observed rank on a fixed collection of questions is known exactly, whereas the rank of a model’s *expected* performance over a broader question population is estimated. MMLU and MMLU-Pro are prominent examples of multi-domain evaluation suites whose finite item banks are used to make broad comparative claims about model capability [Hendrycks et al., 2021, Wang et al., 2024].

The inferential problem does not arise merely because an LLM may generate responses stochastically. From a statistical perspective, uncertainty arises because the 2,000 evaluated questions are treated as a finite sample from a target population of comparable questions. Although this may seem counterintuitive, it implies that rankings computed from any finite set of questions need not coincide with their population-level counterparts. This remains true even when the evaluation uses the full available question bank, since the full benchmark itself is still only a finite realization of a broader population of possible questions¹.

There are two statistical features of our inference problem at hand. First, all models answer the same questions. Let $\hat{\theta}_j$ and $\hat{\theta}_k$ be the accuracies of models j and k . Their difference has variance

$$\text{Var}(\hat{\theta}_j - \hat{\theta}_k) = \text{Var}(\hat{\theta}_j) + \text{Var}(\hat{\theta}_k) - 2 \text{Cov}(\hat{\theta}_j, \hat{\theta}_k). \quad (1)$$

Easy and difficult items tend to move several models together, so the covariance term is generally nonzero. Second, a rank depends on the signs of many pairwise differences and is discontinuous at ties. Directly recomputing ranks under

¹Although on larger benchmarks, the standard error of an accuracy estimate decreases at the usual $(n^{-1/2})$ rate, and many practically meaningful performance gaps might be resolved with high precision.

bootstrap resamples and taking rank percentiles can therefore fail near equal performance [Hall and Miller, 2009, Xie et al., 2009, Mogstad et al., 2024].

We apply the rank-confidence-set construction of Mogstad et al. [2024] to a shared-item LLM benchmark and estimate the full covariance of the score vector from per-question correctness. Our contribution is an empirical and methodological instantiation rather than a new generic theory of ranks. The paper makes three points. First, an exact point ordering overstates what the benchmark resolves. Second, obtaining valid rank statements requires correcting for multiple comparisons and accounting for the dependence across accuracy estimates induced by shared items. Third, rank sets must be interpreted relative to a clearly specified source of randomness.

Our MMLU-Pro study finds unresolved ordering among the three Qwen3 variants near the top and among GLM-4.7-Flash, Llama-3.3-70B, and Mistral-Small-3.2-24B in the middle. The dependence-aware simultaneous analysis identifies a four-model candidate set for the top three.

The remainder of this report is organized as follows: Section 2 reviews related work. Section 3 details our statistical method. Section 4 specifies the experimental setup and presents our main results. Finally, Section 5 discusses the limitations and concludes the report.

2 Related Work

Benchmarking and statistical comparison. MMLU evaluates broad multi-task knowledge, while MMLU-Pro increases difficulty, uses ten answer options, and emphasizes more demanding reasoning questions across 14 domains [Hendrycks et al., 2021, Wang et al., 2024]. Statistical comparison of systems has long emphasized paired evaluation when methods are tested on the same examples: McNemar’s test uses the discordant outcomes of two classifiers rather than treating their accuracies as independent [McNemar, 1947]. In NLP, Dror et al. [2018] survey significance tests and stress that the test must match the data and experimental design. More broadly, benchmark conclusions can vary with data splits, training randomness, and experimental choices [Bouthillier et al., 2021]. These works motivate uncertainty reporting but do not directly turn a collection of pairwise comparisons into confidence sets for ranks.

Inference for ranks. Uncertainty in institutional and performance league tables has been studied for decades [Goldstein and Spiegelhalter, 1996]. Rank is a non-smooth parameter, and naive bootstrap ranks can behave poorly when population features are close [Hall and Miller, 2009]. Smoothed estimators offer one response but require tuning choices [Xie et al., 2009]. Simultaneous rank intervals have also been developed through joint testing and partitioning principles [Al Mohamad et al., 2021]. Most directly, Mogstad et al. [2024] construct marginal confidence sets for one rank, simultaneous confidence sets for all ranks, and sets for top-ranked populations by first obtaining simultaneous confidence intervals for pairwise feature differences. Their framework permits dependent estimators; this is the feature we exploit for shared benchmark items.

Uncertainty-aware LLM evaluation and ranking. Recent LLM work considers several distinct meanings of uncertainty. Ye et al. [2024] incorporate model-output uncertainty into benchmark analysis, whereas our target is sampling uncertainty of mean benchmark performance. For human-preference leaderboards, Chatzi et al. [2024] construct rank sets using prediction-powered inference, and Ameli et al. [2025] model ties and covariance in pairwise chatbot comparisons. Neuhof and Benjamini [2026] develop hierarchical task-level and leaderboard-level rank intervals. These settings involve preferences or aggregation across tasks. We instead consider a fixed-answer benchmark in which every model receives a binary score on every shared item, making the joint accuracy covariance directly estimable.

3 Statistical Framework

3.1 Estimand and observed scores

Suppose p models are evaluated on the same n questions. For item i , define the vector

$$\mathbf{X}_i = (X_{i1}, \dots, X_{ip})^\top, \quad X_{ij} \in \{0, 1\}, \quad (2)$$

where $X_{ij} = 1$ if model j answers item i correctly. We interpret the item vectors as draws from a target distribution over questions and define

$$\boldsymbol{\theta} = \mathbb{E}[\mathbf{X}_i], \quad \hat{\boldsymbol{\theta}} = \overline{\mathbf{X}} = \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i. \quad (3)$$

Thus θ_j is model j 's expected accuracy under the target question distribution, and $\hat{\theta}_j$ is its observed benchmark accuracy. We use the competition-rank convention

$$r_j = 1 + \sum_{k \neq j} \mathbf{1}\{\theta_k > \theta_j\}, \quad (4)$$

so models with equal expected accuracy share a rank.

3.2 Full covariance from shared items

The covariance matrix of the mean vector is estimated by

$$\hat{\boldsymbol{\Sigma}} = \frac{1}{n(n-1)} \sum_{i=1}^n (\mathbf{X}_i - \overline{\mathbf{X}})(\mathbf{X}_i - \overline{\mathbf{X}})^\top. \quad (5)$$

For pair (j, k) , let $\Delta_{jk} = \theta_j - \theta_k$ and $\hat{\Delta}_{jk} = \hat{\theta}_j - \hat{\theta}_k$. The estimated standard error of the paired difference is

$$\widehat{\text{se}}_{jk} = \left(\hat{\boldsymbol{\Sigma}}_{jj} + \hat{\boldsymbol{\Sigma}}_{kk} - 2\hat{\boldsymbol{\Sigma}}_{jk} \right)^{1/2}. \quad (6)$$

If one assumes independence and replaces $\hat{\boldsymbol{\Sigma}}_{jk}$ by zero, this inflates the variance of differences and loses resolving power. Indeed, accounting for shared-item covariance substantially improves resolution relative to a dependence-ignorant analysis: simultaneous singleton ranks increase from 2/13 to 4/13, while average rank-set width falls from 3.31 to 2.38.

3.3 Marginal and simultaneous rank confidence sets

First, we distinguish between a marginal two-sided confidence set for the rank of a particular model j and a simultaneous confidence set. The former guarantees a coverage rate of $1 - \alpha$ for a single model's rank relative to others, whereas the latter provides a joint guarantee that the rank confidence sets for all models hold simultaneously at the $1 - \alpha$ level.

We use a Gaussian parametric bootstrap of the score-estimation error. For draws $b = 1, \dots, B$, sample

$$\mathbf{Z}^{(b)} \sim N(\mathbf{0}, \hat{\boldsymbol{\Sigma}}), \quad \mathbf{Z}^{(b)} = (Z_1^{(b)}, \dots, Z_p^{(b)}). \quad (7)$$

This simulation does not rerun any model. For a pre-specified target model j , define

$$T_j^{(b)} = \max_{k: k \neq j} \frac{|Z_j^{(b)} - Z_k^{(b)}|}{\widehat{\text{se}}_{jk}}, \quad (8)$$

and let $c_j(1 - \alpha)$ be its empirical $(1 - \alpha)$ quantile. For simultaneous inference on the whole leaderboard, use one global maximum

$$T_{\text{all}}^{(b)} = \max_{(j,k): j \neq k} \frac{|Z_j^{(b)} - Z_k^{(b)}|}{\widehat{\text{se}}_{jk}}, \quad (9)$$

with empirical quantile $c_{\text{all}}(1 - \alpha)$. The resulting pairwise interval is

$$C_{jk}(c) = \left[\hat{\Delta}_{jk} - c\widehat{\text{se}}_{jk}, \hat{\Delta}_{jk} + c\widehat{\text{se}}_{jk} \right]. \quad (10)$$

For model j , define

$$N_j^- = \{k \neq j : C_{jk}(c) \subset (-\infty, 0)\}, \quad (11)$$

$$N_j^+ = \{k \neq j : C_{jk}(c) \subset (0, \infty)\}. \quad (12)$$

Algorithm 1 Dependence-aware rank confidence sets from shared benchmark items

Require: Score matrix $X \in \{0, 1\}^{n \times p}$, level α , draws B , mode $m \in \{\text{marginal}, \text{simultaneous}\}$.

```
1: Compute  $\hat{\theta} = n^{-1} \sum_i X_i$  and  $\hat{\Sigma}$  using Equation (5).
2: For every  $j < k$ , compute  $\hat{se}_{jk}$  using Equation (6).
3: for  $b = 1, \dots, B$  do
4:   Draw  $Z^{(b)} \sim N(0, \hat{\Sigma})$ .
5:   if  $m = \text{marginal}$  then
6:     for  $j = 1, \dots, p$  do
7:        $T_j^{(b)} \leftarrow \max_{k \neq j} |Z_j^{(b)} - Z_k^{(b)}| / \hat{se}_{jk}$ .
8:     end for
9:   else
10:     $T_{\text{all}}^{(b)} \leftarrow \max_{(j,k)} |Z_j^{(b)} - Z_k^{(b)}| / \hat{se}_{jk}$ .
11:  end if
12: end for
13: Set  $c_j$  to the  $(1 - \alpha)$  quantile of  $\{T_j^{(b)}\}_b$ , or set one global  $c_{\text{all}}$  from  $\{T_{\text{all}}^{(b)}\}_b$ .
14: for  $j = 1, \dots, p$  do
15:   Form all intervals  $C_{jk}(c)$  using Equation (10), with  $c = c_j$  if  $m = \text{marginal}$  and  $c = c_{\text{all}}$  if  $m = \text{simultaneous}$ .
16:   Determine  $N_j^-$  and  $N_j^+$  via Equations (11) and (12).
17:    $\mathcal{R}_j(c) \leftarrow \{|N_j^-| + 1, \dots, p - |N_j^+|\}$ .
18: end for
19: return  $\{\mathcal{R}_j(c)\}_{j=1}^p$ 
```

Members of N_j^- are established to be better than j , and members of N_j^+ are established to be worse. The rank confidence set is the integer interval

$$\mathcal{R}_j(c) = \{|N_j^-| + 1, \dots, p - |N_j^+|\}. \quad (13)$$

Using c_j gives marginal $1 - \alpha$ coverage for a pre-specified model; using c_{all} for every model gives joint coverage for the complete collection of ranks under the regularity conditions of Mogstad et al. [2024]. The global set is typically wider because it protects all pairwise directional statements simultaneously. Algorithm 1 summarizes the procedure.

4 Results

4.1 Experimental Setup

We analyze 13 LLMs evaluated on a proportional stratified sample of $n = 2,000$ MMLU-Pro questions spanning all 14 subjects. Every model was scored on the same item set, yielding a $2,000 \times 13$ binary correctness matrix. We report raw accuracy; monotone transformations do not change the population ordering and do not create additional pairwise information.

For descriptive score uncertainty, we report two-sided 95% Wilson intervals [Wilson, 1927]. For rank inference, we use $B = 2,000$ ² Gaussian draws and $\alpha = 0.05$. We compare three inferential choices, with marginal and simultaneous variants where applicable:

1. **Naive:** diagonal pairwise standard errors and a fixed 1.96 critical value, with no correction over comparisons.
2. **MRSW-independent:** the max-statistic correction, but with all off-diagonal covariances set to zero.
3. **MRSW-full:** the max-statistic correction and the full shared-item covariance in Equation (5).

The first comparison isolates the effect of multiplicity; the second isolates the information gained from pairing. Across all model pairs, the mean sample correlation of per-item correctness is 0.396.

²We report an additional robustness check in the appendix, varying the number of bootstrap draws. The results are largely unchanged: the simultaneous rank sets remain stable for almost all models, with only a minor change affecting the two top-ranked models. See the appendix for further details.

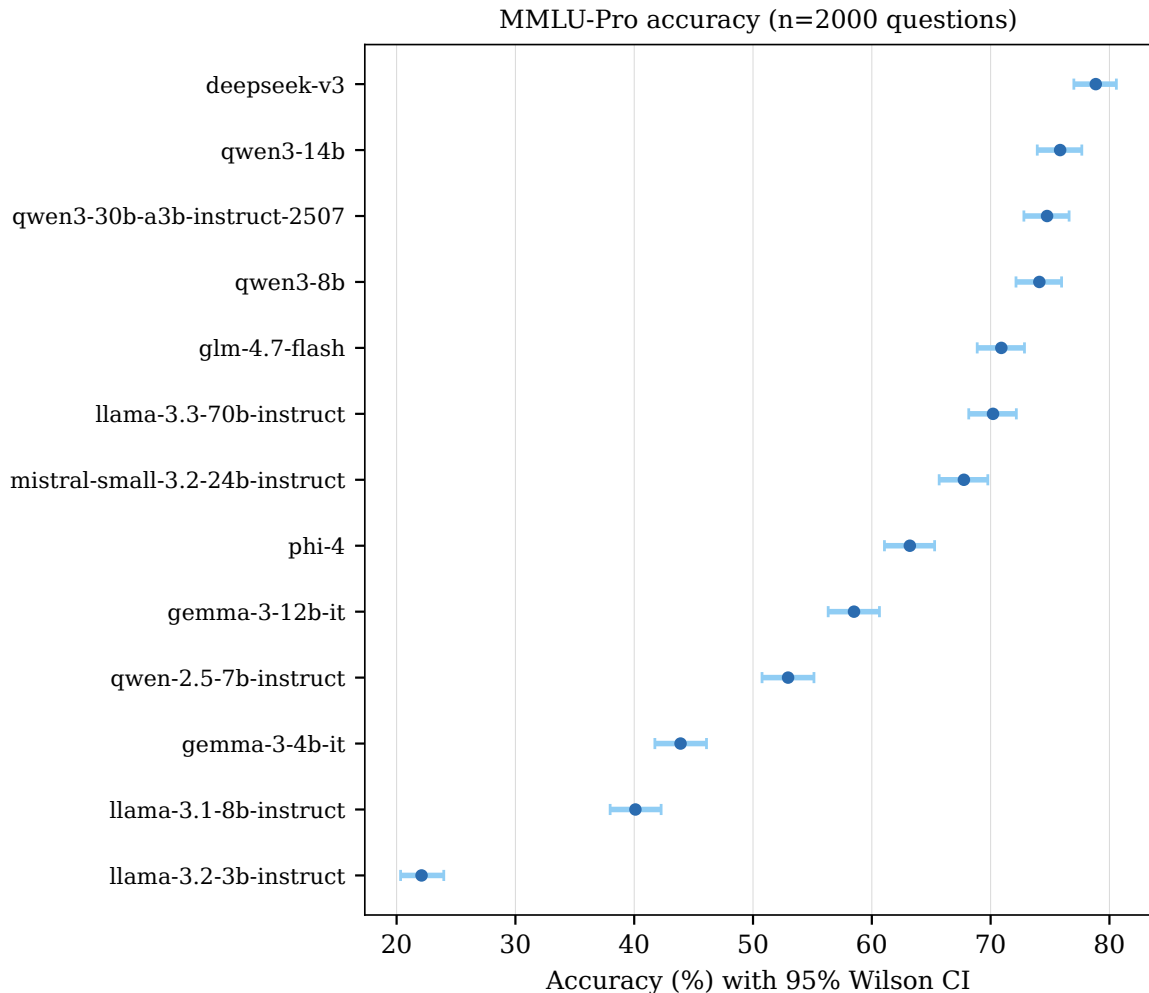


Figure 1: MMLU-Pro accuracies for 13 models on 2,000 questions. Bars are 95% Wilson intervals for each marginal accuracy. These intervals are descriptive and are not themselves used to construct rank confidence sets.

4.2 Accuracy estimates

Figure 1 shows accuracy and Wilson intervals. Scores range from 78.8% for DeepSeek V3 to 22.1% for Llama-3.2-3B. The upper portion of the table is much denser: the three Qwen3 variants score 75.8%, 74.8%, and 74.1%, and GLM-4.7-Flash and Llama-3.3-70B differ by only 0.7 percentage points. Marginal accuracy intervals are useful descriptions of each score, but their visual overlap or non-overlap is not a valid confidence procedure for ranks because each rank depends on multiple paired differences.

4.3 Marginal and simultaneous rank sets

Figure 2 and Table 1 report the main result. Marginal sets answer a question about one pre-specified model at a time. Simultaneous sets support reading the whole leaderboard while maintaining joint 95% coverage.

DeepSeek V3 is assigned marginal set $\{1\}$, but its simultaneous set is $\{1, 2\}$; a whole-leaderboard statement cannot rule out Qwen3-14B at rank one. The ordering among Qwen3-14B, Qwen3-30B-A3B, and Qwen3-8B is unresolved at the reported confidence level: their marginal sets span ranks 2–4, and their simultaneous sets overlap more broadly. This should not be read as proof that their expected accuracies are equal; it means the experiment does not establish a complete ordering among them.

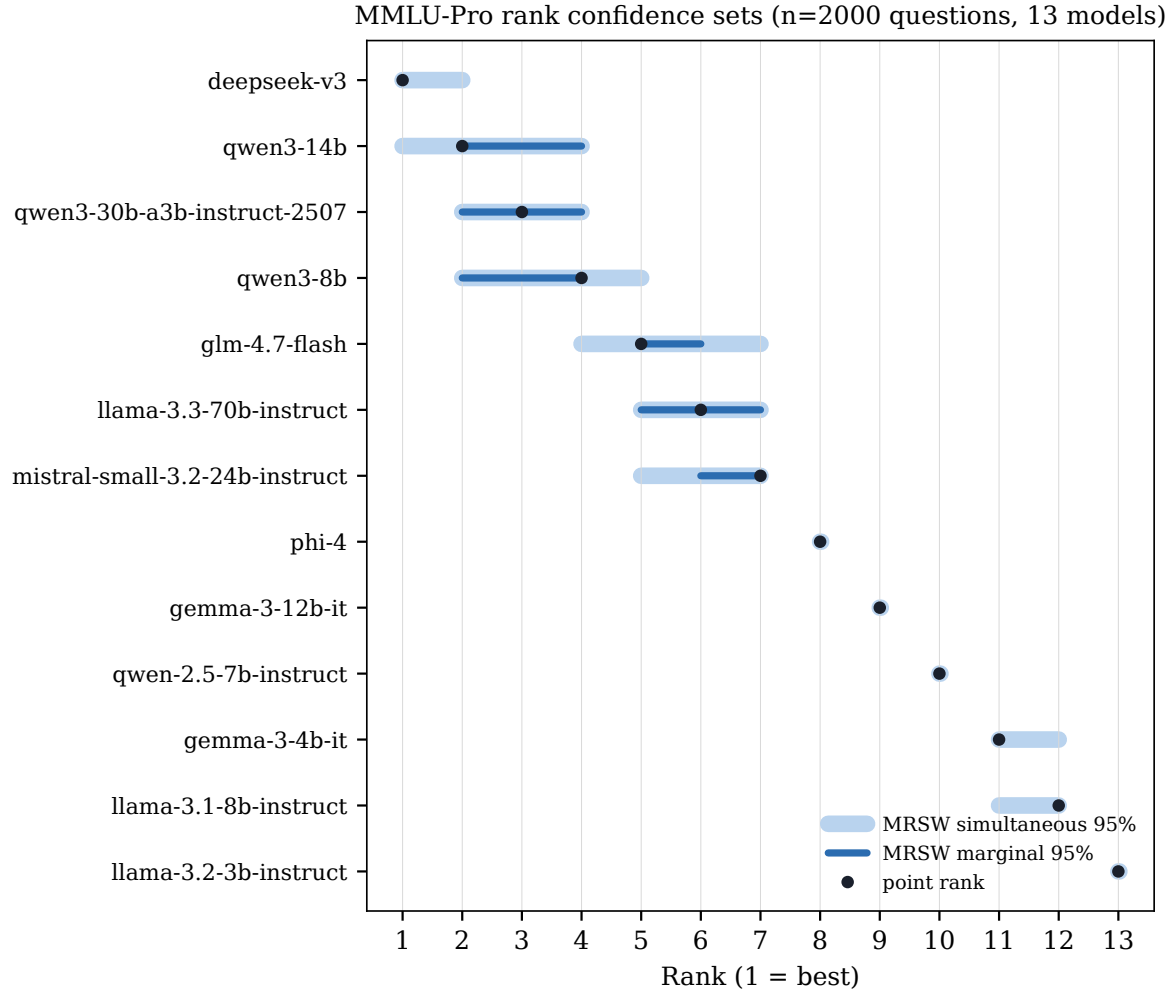


Figure 2: Point ranks (dots), marginal 95% rank confidence sets (dark segments), and simultaneous 95% rank confidence sets (light segments). Rank 1 is best.

A second unresolved group appears around ranks 5–7. By contrast, Phi-4, Gemma-3-12B, and Qwen2.5-7B retain singleton simultaneous ranks, not because their accuracy is measured more precisely (their Wilson CIs are in fact among the widest, being nearest 50%), but because they sit in sparse regions of the accuracy axis: the gap to both neighbours far exceeds the pairwise-difference resolution margin, so every neighbouring comparison is resolved. At the lower end, Gemma-3-4B and Llama-3.1-8B can exchange ranks 11 and 12 under simultaneous inference.

4.4 Multiplicity and shared-item dependence

Table 2 separates two effects that are easy to conflate. Starting from the naive rule, adding the max-statistic correction while continuing to assume independence widens the marginal sets from average size 1.77 to 2.54 and reduces singleton ranks from seven to four. This is the cost of making all comparisons involving a model jointly valid. For the simultaneous leaderboard, the independence analysis has only two singleton ranks and average width 3.31.

Restoring the full covariance recovers much of that lost resolution. In this dataset, the shared-item covariances are mostly positive, so ignoring them tends to overstate pairwise standard errors. Under full covariance, the simultaneous procedure yields four singleton ranks and average width 2.38. The naive and full-covariance marginal methods happen to share the same aggregate width and singleton count, but this is error cancellation, not equivalence: the naive method commits two offsetting errors — assuming independence overestimates the pairwise standard error (widening its sets),

Table 1: MMLU-Pro point accuracies and dependence-aware 95% rank confidence sets. Marginal sets cover one pre-specified model; simultaneous sets cover the full collection of ranks jointly.

Point rank	Model	Acc. (%)	Marginal rank set	Simultaneous rank set
1	DeepSeek V3	78.8	1	1–2
2	Qwen3-14B	75.8	2–4	1–4
3	Qwen3-30B-A3B	74.8	2–4	2–4
4	Qwen3-8B	74.1	2–4	2–5
5	GLM-4.7-Flash	70.9	5–6	4–7
6	Llama-3.3-70B	70.2	5–7	5–7
7	Mistral-Small-3.2-24B	67.8	6–7	5–7
8	Phi-4	63.2	8	8
9	Gemma-3-12B	58.5	9	9
10	Qwen2.5-7B	52.9	10	10
11	Gemma-3-4B	43.9	11	11–12
12	Llama-3.1-8B	40.1	12	11–12
13	Llama-3.2-3B	22.1	13	13

while omitting the multiple-comparison correction understates the critical value (narrowing them). The two biases cancel here, so it coincidentally lands on the values that the full-covariance method reaches legitimately, by applying a correctly larger bootstrapped critical value to a correctly smaller, correlation-adjusted standard error.

Table 2: Resolving power under alternative treatment of multiplicity and dependence. Width is the number of integer ranks in a model’s confidence set, averaged across 13 models.

Method	Singleton ranks	Mean width
Naive, independent, no multiplicity correction	7/13	1.77
MRSW-independent, marginal	4/13	2.54
MRSW-independent, simultaneous	2/13	3.31
MRSW-full, marginal	7/13	1.77
MRSW-full, simultaneous	4/13	2.38

4.5 Top- and bottom-tier candidates

Projecting the simultaneous rank sets onto tier membership gives a 95% jointly valid candidate set. Four models cannot be excluded from the top three: DeepSeek V3 and the three Qwen3 variants. Thus, the point leaderboard’s top three is more specific than the evidence supports. The bottom three are resolved exactly as a set: Gemma-3-4B, Llama-3.1-8B, and Llama-3.2-3B. These are candidate-set statements, not claims that every included model is definitely in the tier.

5 Conclusion

5.1 Discussion and Limitations

What they do not capture. The experiment has one realized answer per model-question pair. Consequently, it does not separately estimate variability from decoding seeds, prompt templates, judge randomness, or model-provider updates. Benchmark contamination, annotation error, and changes to the candidate model set are also outside the coverage statement. These factors may dominate item-sampling uncertainty in some deployments and require repeated or redesigned evaluations rather than a different transformation of accuracy.

Sampling design. The current covariance treats evaluated items as draws from the benchmark mixture. Because the sample was selected proportionally within 14 subjects, a design-based analysis conditioning on fixed subject counts could instead combine within-subject covariance estimates. That alternative is a useful robustness check.

Non-resolution is not equivalence. An interval spanning several ranks says that the data do not establish all relevant directional comparisons at the chosen error level. It does not prove practical equivalence. If an application requires a minimum meaningful accuracy gap, that margin should be specified before analysis and incorporated into the pairwise hypotheses.

5.2 Concluding Remarks

A point leaderboard reports more ordering information than a finite shared-item benchmark may support. Applying rank confidence sets to 13 models on 2,000 MMLU-Pro questions reveals that several neighboring point ranks are unresolved, especially near the top. Valid multiplicity control widens the ranking, while the positive dependence induced by shared questions restores substantial power when modeled through the full covariance matrix. The resulting simultaneous intervals give a direct answer to the leaderboard-level question: which ranks can be asserted jointly from this benchmark sample? They should be reported alongside point accuracies and an explicit statement that the uncertainty target is item sampling, not every source of LLM evaluation variability.

Future research could extend the present framework in several directions. First, it would be useful to examine whether the rank-confidence sets are sensitive to alternative resampling schemes, such as a nonparametric bootstrap over benchmark items. Unlike many social-science applications, benchmark evaluations often provide item-level information on each model’s response and correctness, making it possible to resample directly from the observed evaluation data rather than relying only on aggregate accuracy estimates.

Second, the method has a natural practical application in model selection. For a given task or domain, one could design a targeted problem set and use simultaneous rank-confidence sets to determine whether competing models are statistically distinguishable. If two models are ranked similarly and their confidence sets overlap substantially, users may prefer the more cost-effective model for deployment rather than the model with the slightly higher point estimate.

Finally, this framework could be used to study the evolution of model capabilities over time. By applying the same rank-inference procedure across model generations, one could identify where newer models make statistically meaningful improvements and which components of performance contribute most to changes in their relative ranking.

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6 Appendix

6.1 Additional Robustness Check

Sensitivity of the 95% simultaneous rank-confidence sets to the number of bootstrap draws R . For each R , $\bar{c}_{\text{sim}}$ denotes the simultaneous critical value averaged over 60 independent bootstrap seeds, with the Monte Carlo standard error of the mean reported in parentheses. This estimator is consistent for the limiting value c_{∞} , and its Monte Carlo standard deviation decreases at rate $R^{-1/2}$. The simultaneous rank sets are invariant to R for 11 of the 13 models. The only exception is the leading pair, deepseek-v3 and qwen3-14b, whose relative order is determined by the standardized accuracy gap

$$T = \frac{\hat{\theta}_1 - \hat{\theta}_2}{\widehat{\text{se}}} = 3.3111,$$

which is a fixed function of the data. A high-draw approximation gives limiting critical value $c_{\infty} = 3.3000$, so that $T - c_{\infty} = 0.0111$, approximately 18 Monte Carlo standard errors. Thus, as the number of bootstrap draws tends to infinity, deepseek-v3 attains rank 1 at the 5% simultaneous level. However, this conclusion is close to the decision boundary, since the standardized gap exceeds the limiting critical value by only 0.0111. The column $\hat{P}(>)$ reports the empirical fraction of the 60 seeds for which deepseek-v3 is ranked strictly above qwen3-14b. The rank-set columns are evaluated at the canonical seed 0. The final column reports the number of the 13 model-level rank sets that differ from the $R = 2000$ specification; only the rank sets of the two leading models ever change.

Table 3: **Robustness of the simultaneous rank sets to the bootstrap draw count R .**

R	$\bar{c}_{\text{sim}}$ (SE)	$\hat{P}(\succ)$	deepseek-v3	qwen3-14b	# sets $\neq R=2000$
2,000	3.2936 (0.0035)	0.77	{1-2}	{1-4}	0/13
5,000	3.2966 (0.0029)	0.75	{1-2}	{1-4}	0/13
10,000	3.3030 (0.0020)	0.75	{1-2}	{1-4}	0/13
20,000	3.3017 (0.0014)	0.77	{1-2}	{1-4}	0/13
50,000	3.2992 (0.0008)	0.98	{1}	{2-4}	2/13
100,000	3.2995 (0.0006)	1.00	{1}	{2-4}	2/13
200,000	3.3001 (0.0004)	1.00	{1}	{2-4}	2/13

6.2 Full Accuracy and Rank Table

Table 4: Full numerical results. Accuracy intervals are 95% Wilson intervals. Rank sets use the full shared-item covariance.

Rank	Model identifier	Acc.	Wilson CI	Marginal	Simultaneous
1	DeepSeek V3	78.8	[77.0, 80.6]	1	1-2
2	qwen3-14b	75.8	[73.9, 77.7]	2-4	1-4
3	qwen3-30b-a3b-instruct-2507	74.8	[72.8, 76.6]	2-4	2-4
4	qwen3-8b	74.1	[72.1, 76.0]	2-4	2-5
5	glm-4.7-flash	70.9	[68.9, 72.8]	5-6	4-7
6	llama-3.3-70b-instruct	70.2	[68.2, 72.2]	5-7	5-7
7	mistral-small-3.2-24b-instruct	67.8	[65.7, 69.8]	6-7	5-7
8	phi-4	63.2	[61.1, 65.3]	8	8
9	gemma-3-12b-it	58.5	[56.3, 60.6]	9	9
10	qwen-2.5-7b-instruct	52.9	[50.8, 55.1]	10	10
11	gemma-3-4b-it	43.9	[41.7, 46.1]	11	11-12
12	llama-3.1-8b-instruct	40.1	[38.0, 42.3]	12	11-12
13	llama-3.2-3b-instruct	22.1	[20.3, 24.0]	13	13

6.3 Projected Top-3 and Bottom-3 Candidate Sets

For a simultaneous rank interval $[L_j, U_j]$, the projected top- τ candidate set contains models with $L_j \leq \tau$, and the projected bottom- τ set contains models with $U_j \geq p - \tau + 1$. For $\tau = 3$, the results are:

Table 5: Models that cannot be excluded from the top three or bottom three under the simultaneous rank sets.

Tier	Model	Simultaneous rank set
Top-3	DeepSeek V3	1-2
	Qwen3-14B	1-4
	Qwen3-30B-A3B	2-4
	Qwen3-8B	2-5
Bottom-3	Gemma-3-4B	11-12
	Llama-3.1-8B	11-12
	Llama-3.2-3B	13